

100 Docker Troubleshooting Scenarios

Real-World Diagnosis & Fix Guide

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CONTAINER LIFECYCLE — Real-World Troubleshooting

#1 Container Exits Immediately After Start

SYMPTOM	docker run completes instantly; container shows Exited (0) or Exited (1) right away.
ROOT CAUSE	No foreground process to keep container alive, or entrypoint command fails immediately.
FIX	Use a foreground CMD (e.g., tail -f /dev/null for debug). Check logs: docker logs <container>.
<pre>docker logs <container_id></pre>	

#2 Container Stuck in Restarting Loop

SYMPTOM	docker ps shows container perpetually restarting with short uptime.
ROOT CAUSE	App crash on startup, bad entrypoint, or missing dependency (env var, file, network).
FIX	Inspect logs between restarts. Set --restart=no temporarily to prevent loop while debugging.
<pre>docker logs --tail=50 <container></pre>	

#3 Container Won't Stop — docker stop Hangs

SYMPTOM	docker stop takes 10 seconds and force-kills; app state may be lost.
ROOT CAUSE	Application not handling SIGTERM signal; no graceful shutdown logic.
FIX	Implement SIGTERM handler in app. Use CMD not ENTRYPOINT exec form. Lower --time if needed.
<pre>docker stop --time=5 <container></pre>	

#4 Container Removed Unexpectedly

SYMPTOM	Running container disappears; docker ps shows nothing; no crash logs.
ROOT CAUSE	Container started without --restart policy and host rebooted, or OOM kill by kernel.
FIX	Use --restart=unless-stopped. Check dmesg for OOM events. Set memory limits.
<pre>dmesg grep -i 'oom\ killed'</pre>	

#5 docker run Returns Permission Denied

SYMPTOM	docker: Got permission denied while trying to connect to the Docker daemon socket.
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ROOT CAUSE	Current user not in docker group; Docker daemon not running as expected user.
FIX	Add user to docker group: <code>sudo usermod -aG docker \$USER</code> . Re-login or newgrp docker.
	<code>sudo usermod -aG docker \$USER && newgrp docker</code>

#6 Container Running but Process Zombie

SYMPTOM	ps aux inside container shows zombie (Z) processes accumulating over time.
ROOT CAUSE	No init process; child processes not reaped after parent exits (PID 1 not handling signals).
FIX	Use --init flag to add tini as PID 1 init process that reaps zombies.
	<code>docker run --init my-image</code>

#7 Container Timezone Mismatch

SYMPTOM	Application logs show wrong timestamp; container time differs from host.
ROOT CAUSE	Container uses UTC by default regardless of host timezone.
FIX	Mount host timezone: <code>-v /etc/localtime:/etc/localtime:ro</code> or set TZ env variable.
	<code>docker run -e TZ=Asia/Kolkata -v /etc/localtime:/etc/localtime:ro my-image</code>

#8 docker exec Fails — Container Not Running

SYMPTOM	docker exec returns: Error: No such container or container not running.
ROOT CAUSE	Container has stopped or exited; exec only works on running containers.
FIX	Use docker start <container> first. Or use <code>docker run --rm -it <image> sh</code> for one-off debug.
	<code>docker start <container> && docker exec -it <container> sh</code>

#9 Entrypoint Script Not Executable

SYMPTOM	Container exits with: <code>exec /entrypoint.sh: permission denied</code> .
ROOT CAUSE	COPY adds script without execute permission; or Windows line endings (CRLF) in script.
FIX	Add <code>RUN chmod +x /entrypoint.sh</code> in Dockerfile. Use dos2unix for CRLF issues.
	<code>RUN chmod +x /entrypoint.sh</code>

#10 Container Cannot Access Host Services

SYMPTOM	Container cannot connect to service running on host (e.g., localhost:5432).
ROOT CAUSE	Container network is isolated; localhost inside container refers to container, not host.
FIX	Use host.docker.internal (Mac/Win) or --add-host=host.docker.internal:host-gateway (Linux).
<pre>docker run --add-host=host.docker.internal:host-gateway my-image</pre>	

#11 Environment Variables Not Available in Container

SYMPTOM	Application inside container cannot read expected env vars; they are empty.
ROOT CAUSE	Vars passed with -e but typo in name, or .env file not loaded.
FIX	Verify: docker exec <c> env. Use --env-file=.env for file-based injection.
<pre>docker run --env-file=.env my-image</pre>	

#12 Container Hostname Not Resolving Between Containers

SYMPTOM	Container A cannot ping or curl container B by hostname.
ROOT CAUSE	Containers on different networks, or not using Docker Compose service names as hostnames.
FIX	Attach both containers to same user-defined bridge network. Default bridge doesn't support DNS.
<pre>docker network connect my-net containerA && docker network connect my-net containerB</pre>	

#13 Container Working Directory Not Set Correctly

SYMPTOM	App fails to find relative-path files; pwd inside container is wrong directory.
ROOT CAUSE	WORKDIR not set in Dockerfile; defaults to /.
FIX	Add WORKDIR /app in Dockerfile before CMD/ENTRYPOINT.
<pre>WORKDIR /app</pre>	

#14 Container User Running as Root Unexpectedly

SYMPTOM	Security scan flags container running as root; compliance check fails.
ROOT CAUSE	No USER instruction in Dockerfile; default is root.
FIX	Add USER nonroot after creating the user. Use --user flag at runtime if needed.
<pre>RUN adduser -D appuser && USER appuser</pre>	

#15 Shared Memory Size Too Small

SYMPTOM	Application crashes with: No space left on device on /dev/shm.
ROOT CAUSE	Default Docker shm size is 64MB; some apps (Chrome, Postgres) need more.
FIX	Increase with --shm-size flag at runtime.
<pre>docker run --shm-size=256m my-image</pre>	

IMAGE ISSUES — Real-World Troubleshooting

#16 Image Pull Rate Limit Hit (Docker Hub)

SYMPTOM	docker pull fails: toomanyrequests: You have reached your pull rate limit.
ROOT CAUSE	Anonymous Docker Hub pulls limited to 100/6h; authenticated to 200/6h.
FIX	docker login to authenticate. Use a mirror or pull-through cache. Move to private registry.
	<pre>docker login && docker pull my-image:tag</pre>

#17 Image Not Found — Correct Tag Required

SYMPTOM	Unable to find image 'myapp:latest' locally followed by pull error.
ROOT CAUSE	Image tag doesn't exist on registry; typo in image name or tag.
FIX	docker search or check registry UI for correct tag. Use digest for immutable reference.
	<pre>docker pull myrepo/myapp@sha256:<digest></pre>

#18 Large Image Size Slowing CI/CD

SYMPTOM	docker build and push takes many minutes; image is 1GB+.
ROOT CAUSE	Using full OS base image, installing dev tools, not using multi-stage builds, or not cleaning apt cache.
FIX	Use alpine/distroless base. Use multi-stage builds. RUN apt-get install ... && rm -rf /var/lib/apt/lists/*
	<pre>docker image ls --format '{{.Repository}}:{{.Tag}}\t{{.Size}}'</pre>

#19 Docker Build Cache Invalidated Unexpectedly

SYMPTOM	Every docker build runs all layers from scratch; caching not working.
ROOT CAUSE	COPY . . too early in Dockerfile invalidates cache on any file change.
FIX	Copy dependency files first (package.json, requirements.txt), install, then COPY rest of source.
	<pre>COPY package*.json ./ \nRUN npm install \nCOPY . .</pre>

#20 Multi-Stage Build Final Image Missing Files

SYMPTOM	Container starts but required binaries or config files are absent.
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ROOT CAUSE	COPY --from=builder stage missing correct source path, or wrong stage name referenced.
FIX	Verify paths with docker run --rm builder-stage ls /app. Check COPY --from stage names.
<pre>docker build --target builder -t debug-build . && docker run --rm debug-build ls -la /app</pre>	

#21 Image Layers Not Cleaned Up — Disk Full

SYMPTOM	docker build fails: no space left on device; /var/lib/docker consuming all disk.
ROOT CAUSE	Dangling images, stopped containers, and unused volumes accumulating over time.
FIX	docker system prune -af --volumes. Set up automated cleanup cron job.
<pre>docker system prune -af --volumes && docker system df</pre>	

#22 Wrong Architecture Image (ARM vs AMD64)

SYMPTOM	exec /app/server: exec format error — container crashes on start.
ROOT CAUSE	Image built for amd64 but running on ARM host (or vice versa).
FIX	Build multi-arch image with docker buildx build --platform linux/amd64,linux/arm64.
<pre>docker buildx build --platform linux/amd64,linux/arm64 -t myapp:multi .</pre>	

#23 Base Image Tag 'latest' Causes Inconsistency

SYMPTOM	Builds work locally but fail in CI; different behavior across machines.
ROOT CAUSE	latest tag points to different versions at different times; non-reproducible builds.
FIX	Always pin to specific digest or version tag. Never use latest in production Dockerfiles.
<pre>FROM node:20.11.0-alpine3.19</pre>	

#24 Secret Baked Into Docker Image Layer

SYMPTOM	Security scan finds API keys or passwords in image history.
ROOT CAUSE	ARG or ENV used for secrets; docker history shows the secret in a layer.
FIX	Use --secret mount with BuildKit. Never pass secrets via ARG/ENV in production.
<pre>docker build --secret id=mysecret,src=./secret.txt -t my-image .</pre>	

#25 Dockerfile RUN Command Fails Silently

SYMPTOM	Build succeeds but expected file or package is missing at runtime.
ROOT CAUSE	Command in RUN used true or redirected stderr; failure was swallowed.
FIX	Remove true. Use RUN set -e && command. Always check exit codes.
	<pre>RUN set -e && apt-get install -y mypackage</pre>

#26 docker build Context Too Large — Slow Build

SYMPTOM	docker build first step (Sending build context) takes very long — sending hundreds of MBs.
ROOT CAUSE	No .dockerignore file; node_modules, .git, or large test files included in context.
FIX	Add .dockerignore with node_modules, .git, *.log, dist/, etc.
	<pre>echo 'node_modules\n.git\ndist\n*.log' > .dockerignore</pre>

#27 Image Digest Mismatch After Push

SYMPTOM	Pushed image digest differs from local build digest.
ROOT CAUSE	Registry transforms or recompresses layers on push, changing digest.
FIX	Use the registry-reported digest for pinning, not local build digest.
	<pre>docker inspect --format='{{index .RepoDigests 0}}' myimage:tag</pre>

#28 Cannot Delete Image — Container Still Using It

SYMPTOM	docker rmi returns: image is being used by stopped container.
ROOT CAUSE	Stopped containers still reference image layers; cannot remove until containers deleted.
FIX	docker rm \$(docker ps -aq) to remove all stopped containers, then rmi.
	<pre>docker rm \$(docker ps -aq) && docker rmi <image_id></pre>

#29 COPY vs ADD Confusion Causing Build Failures

SYMPTOM	ADD with URL doesn't work as expected; or tar not auto-extracted when expected.
ROOT CAUSE	ADD auto-extracts tarballs and fetches URLs (discouraged); COPY is explicit and preferred.
FIX	Use COPY for local files. Use RUN curl + RUN tar for downloads to keep layers explicit.
	<pre>COPY ./app /app</pre>

#30 Layer Count Too High — Slow Pull

SYMPTOM	docker pull takes long; image has 50+ layers.
ROOT CAUSE	Each RUN/COPY/ADD creates a new layer; too many separate RUN commands.
FIX	Chain RUN commands with && to merge into fewer layers.
<pre>RUN apt-get update && apt-get install -y curl git && rm -rf /var/lib/apt/lists/*</pre>	

NETWORKING — Real-World Troubleshooting

#31 Port Already in Use on Host

SYMPTOM	docker run fails: Bind for 0.0.0.0:8080 failed: port is already allocated.
ROOT CAUSE	Another process or Docker container already bound to the host port.
FIX	Use <code>lsof -i :8080</code> to find the process. Change host port mapping: <code>-p 8081:8080</code> .
	<pre>lsof -i :8080 && docker run -p 8081:8080 my-image</pre>

#32 Container Cannot Reach Internet

SYMPTOM	curl google.com inside container fails: Network unreachable or name resolution failure.
ROOT CAUSE	Docker network bridge not connected to internet, or iptables rules missing after restart.
FIX	Check docker network inspect bridge. Restart Docker daemon. Verify iptables/nftables forwarding rules.
	<pre>sudo systemctl restart docker && docker run --rm alpine ping -c3 8.8.8.8</pre>

#33 DNS Resolution Failing Inside Container

SYMPTOM	curl http://myservice fails with: could not resolve host.
ROOT CAUSE	Container DNS configured incorrectly; default 8.8.8.8 blocked by corporate firewall.
FIX	Set custom DNS: <code>--dns 192.168.1.1</code> . Or configure daemon.json with dns entry.
	<pre>echo '{"dns": ["8.8.8.8", "1.1.1.1"]}' sudo tee /etc/docker/daemon.json && sudo systemctl restart docker</pre>

#34 Container IP Conflicts with Host Network

SYMPTOM	Containers cannot communicate; host routing broken after Docker network creation.
ROOT CAUSE	Docker default bridge subnet (172.17.0.0/16) conflicts with host or VPN subnet.
FIX	Change default-address-pools in daemon.json. Create custom networks with non-conflicting subnets.
	<pre>docker network create --subnet=10.10.0.0/24 my-custom-net</pre>

#35 --network=host Not Working on Docker Desktop (Mac/Windows)

SYMPTOM	docker run --network=host but container still cannot access host services on Mac/Win.
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ROOT CAUSE	Docker Desktop runs Docker inside a VM; host networking refers to VM, not Mac/Win host.
FIX	Use host.docker.internal to reach services on the actual Mac/Windows host machine.
	<code>docker run -e DB_HOST=host.docker.internal my-image</code>

#36 Exposed Port Not Accessible from Outside Host

SYMPTOM	Container port mapped to host but requests from other machines time out.
ROOT CAUSE	Host firewall (ufw/iptables) blocking inbound traffic on mapped port.
FIX	Add firewall rule: ufw allow 8080. Or Docker manages iptables — check Docker iptables rules.
	<code>sudo ufw allow 8080/tcp && docker port <container></code>

#37 Container-to-Container Network Latency High

SYMPTOM	Requests between containers on same host have unexpectedly high latency.
ROOT CAUSE	Using default bridge with --link (legacy); or overlay network overhead when not needed.
FIX	Use user-defined bridge networks for same-host communication. Avoid overlay unless multi-host.
	<code>docker network create --driver bridge fast-net</code>

#38 Docker Compose Service Cannot Reach Another Service

SYMPTOM	Service A in Compose cannot curl service B by service name.
ROOT CAUSE	Services on different networks in Compose, or network not explicitly defined.
FIX	Define a shared networks: section and assign both services. Or rely on Compose default network.
	<code>docker compose exec serviceA curl http://serviceB:8080</code>

#39 Overlay Network Not Working in Swarm

SYMPTOM	Services on different Swarm nodes cannot communicate over overlay network.
ROOT CAUSE	Required ports for Swarm (2377, 7946, 4789) blocked by host firewall.
FIX	Open TCP 2377, TCP/UDP 7946, UDP 4789 between all Swarm nodes.
	<code>sudo ufw allow 2377/tcp && sudo ufw allow 7946 && sudo ufw allow 4789/udp</code>

#40 Host-Mode Container Conflicting with Host Ports

SYMPTOM	Container with --network=host fails to start: address already in use.
ROOT CAUSE	App inside host-mode container binds to port already used by host process.
FIX	Find and stop conflicting host process. Or change app port via env variable.
<pre>ss -tulpn grep <port></pre>	

#41 Published Port Not Responding After Host Reboot

SYMPTOM	After host restart, containers running but ports not accessible.
ROOT CAUSE	iptables rules managed by Docker lost after reboot; iptables-persistent not set up.
FIX	Install iptables-persistent. Or ensure Docker starts before iptables restore. Restart Docker.
<pre>sudo systemctl restart docker</pre>	

#42 IPv6 Traffic Not Routed to Container

SYMPTOM	IPv6 requests to container fail even though port is published.
ROOT CAUSE	Docker IPv6 support disabled by default in daemon.
FIX	Enable IPv6 in daemon.json: ipv6: true, fixed-cidr-v6. Restart Docker.
<pre>echo '{"ipv6":true,"fixed-cidr-v6":"fd00::/80"}' sudo tee /etc/docker/daemon.json</pre>	

#43 Too Many Docker Networks Causing Routing Issues

SYMPTOM	docker network ls shows 50+ networks; host networking becomes erratic.
ROOT CAUSE	Each Docker Compose project creates a default network; they accumulate over time.
FIX	docker network prune to remove unused networks. Use explicit network names in Compose.
<pre>docker network prune -f && docker network ls</pre>	

#44 Container Cannot Connect to External Database

SYMPTOM	App container connection refused or timeout to RDS/Cloud DB.
ROOT CAUSE	Security group/firewall not allowing Docker host IP, or wrong DB hostname used.
FIX	Add Docker host's public/private IP to DB security group. Verify DNS resolution inside container.
<pre>docker exec <container> nc -zv <db-host> 5432</pre>	

#45 Macvlan Network Container Not Reachable

SYMPTOM	Container on macvlan network not reachable from host or other containers.
ROOT CAUSE	Macvlan promiscuous mode not enabled on host NIC, or IP conflicts.
FIX	Enable promiscuous: ip link set eth0 promisc on. Ensure container IP is outside DHCP range.
<pre>sudo ip link set eth0 promisc on</pre>	

VOLUMES & STORAGE — Real-World Troubleshooting

#46 Volume Mount Shows Empty Directory

SYMPTOM	Bind mount of host directory into container shows no files inside container.
ROOT CAUSE	Wrong host path, or host path does not exist (Docker creates empty dir automatically).
FIX	Use absolute paths. Verify host path exists before mounting.
<pre>docker run -v \$(pwd)/data:/app/data my-image</pre>	

#47 Named Volume Not Persisting Data

SYMPTOM	Data written to volume disappears after container removed and recreated.
ROOT CAUSE	Using anonymous volume (not named); or container writing to different path than volume mount.
FIX	Use named volumes: -v myvolume:/data. Verify app writes to the mounted path.
<pre>docker volume create myvolume && docker run -v myvolume:/data my-image</pre>	

#48 Permission Denied Writing to Bind Mount

SYMPTOM	App gets permission denied writing to bind-mounted host directory.
ROOT CAUSE	Host directory owned by root or different UID than container user.
FIX	chown host dir to match container UID. Or use --user \$(id -u):\$(id -g) at runtime.
<pre>sudo chown -R 1000:1000 ./data && docker run -v \$(pwd)/data:/app/data my-image</pre>	

#49 Volume Data Not Backed Up

SYMPTOM	Container volume data lost after accidental docker volume rm or host failure.
ROOT CAUSE	No backup strategy for Docker volumes; volumes deleted with --volumes flag.
FIX	Use docker run --rm -v myvolume:/data -v \$(pwd):/backup alpine tar to backup volumes.
<pre>docker run --rm -v myvolume:/data -v \$(pwd):/backup alpine tar czf /backup/vol.tar.gz /data</pre>	

#50 tmpfs Mount Not Available in Older Docker

SYMPTOM	docker run --tmpfs /tmp fails: unknown flag or not supported.
ROOT CAUSE	--tmpfs support added in Docker 1.10; older daemon version in use.

FIX	Upgrade Docker daemon. Or use <code>--mount type=tmpfs,destination=/tmp</code> as alternative.
	<pre>docker run --mount type=tmpfs,destination=/tmp my-image</pre>

#51 Docker Volume on NFS Causes Stale File Handle

SYMPTOM	App errors: stale file handle or input/output error on NFS-backed volume.
ROOT CAUSE	NFS mount has been disconnected or NFS server rebooted; Docker volume still references old handle.
FIX	Remount NFS volume. Use <code>docker volume rm</code> and recreate. Check NFS server health.
	<pre>docker volume rm <vol> && docker volume create --driver local --opt type=nfs --opt o=addr=<nfs-server>,rw --opt device=:/export/data nfs-vol</pre>

#52 Write Performance Slow on Bind Mount (Mac/Windows)

SYMPTOM	Application extremely slow on Docker Desktop with bind-mounted code.
ROOT CAUSE	File system virtualization between host and Docker VM causes high I/O overhead on Mac/Windows.
FIX	Use Docker-managed volumes for data. For code, use <code>:cached</code> or <code>:delegated</code> flag. Enable VirtioFS.
	<pre>docker run -v \$(pwd):/app:cached my-image</pre>

#53 Volume Used by Multiple Containers Causing Corruption

SYMPTOM	Two containers writing to same volume; data corrupted or inconsistent.
ROOT CAUSE	No file locking coordination between containers sharing a volume.
FIX	Designate single writer container. Use volume drivers with locking support for shared write scenarios.
	<pre>docker run -v sharedvol:/data:ro consumer-image</pre>

#54 Cannot Remove Volume — In Use

SYMPTOM	<code>docker volume rm</code> fails: volume is in use by a container.
ROOT CAUSE	Stopped (not removed) containers still hold volume reference.
FIX	<code>docker rm</code> all stopped containers that use the volume, then <code>volume rm</code> .
	<pre>docker ps -a --filter volume=<vol-name> && docker rm <container> && docker volume rm <vol></pre>

#55 Container Log File Growing Unbounded

SYMPTOM	Host disk fills up; /var/lib/docker/containers/<id>/<id>-json.log is gigabytes.
ROOT CAUSE	No log rotation configured; Docker default logging driver writes unbounded.
FIX	Set max-size and max-file in daemon.json or per-container with --log-opt.
<pre>docker run --log-opt max-size=10m --log-opt max-file=3 my-image</pre>	

#56 Overlay2 Storage Driver Out of Space

SYMPTOM	docker build or run fails: no space left on device even though df -h shows free space.
ROOT CAUSE	Docker uses separate filesystem (/var/lib/docker on overlay2); that partition is full.
FIX	Move /var/lib/docker to larger partition. Or run docker system prune.
<pre>docker system df && docker system prune -af</pre>	

#57 SELinux Blocking Volume Mount

SYMPTOM	Permission denied on mounted volume even with correct host ownership; SELinux denials in audit.log.
ROOT CAUSE	SELinux policy preventing container from accessing host path.
FIX	Append :z (shared) or :Z (private) to volume mount for automatic SELinux relabeling.
<pre>docker run -v /host/data:/app/data:Z my-image</pre>	

#58 Volume Driver Plugin Not Found

SYMPTOM	docker volume create fails: Error looking up volume plugin: plugin not found.
ROOT CAUSE	Third-party volume plugin (e.g., rexray, convoy) not installed or not running.
FIX	Install volume plugin: docker plugin install <plugin>. Verify with docker plugin ls.
<pre>docker plugin install rexray/ebs && docker plugin ls</pre>	

DOCKER COMPOSE — Real-World Troubleshooting

#59 Service Dependency Not Ready (depends_on Not Waiting)

SYMPTOM	App container starts before DB is ready; connection refused errors at startup.
ROOT CAUSE	depends_on only waits for container to start, not for service inside to be healthy.
FIX	Add healthcheck to DB service and use depends_on condition: service_healthy.
<pre>depends_on:\n db:\n condition: service_healthy</pre>	

#60 Compose File Variable Substitution Failing

SYMPTOM	\${VARIABLE} in Compose file appears literally; env substitution not working.
ROOT CAUSE	.env file not in same directory as docker-compose.yml, or variable not set.
FIX	Create .env in same directory. Or use --env-file flag. Verify with docker compose config.
<pre>docker compose --env-file .env.prod config</pre>	

#61 Port Conflict When Scaling Services

SYMPTOM	docker compose up --scale app=3 fails: port is already allocated.
ROOT CAUSE	All scaled instances try to bind to same host port.
FIX	Remove host port mapping when scaling; use a load balancer in front. Let Docker assign ephemeral ports.
<pre>docker compose up --scale app=3 -d</pre>	

#62 Compose Network Not Removed After Down

SYMPTOM	docker compose down leaves network behind; subsequent up fails with network already exists.
ROOT CAUSE	External network referenced but not owned by Compose; or network has active endpoints.
FIX	Add --remove-orphans flag. Use docker network rm manually if needed.
<pre>docker compose down --remove-orphans --volumes</pre>	

#63 Volume Not Shared Between Compose Services

SYMPTOM	Service A writes file; service B cannot see it even though both mount same volume.
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ROOT CAUSE	Volume not declared in top-level volumes: section; each service creates separate anonymous volume.
FIX	Declare named volume in top-level volumes: and reference in both services.
	<pre>volumes:\n shared-data:\nservices:\n a:\n volumes: [shared-data:/data]</pre>

#64 Compose Build Ignores Code Changes

SYMPTOM	After code change, docker compose up uses old image without rebuilding.
ROOT CAUSE	Compose uses cached image; --build flag not used.
FIX	Always use docker compose up --build during development to force rebuild.
	<pre>docker compose up --build --force-recreate</pre>

#65 Compose Service Name DNS Not Resolving

SYMPTOM	Service A cannot reach service B using service name as hostname.
ROOT CAUSE	Services on different explicitly-defined networks, or network not attached to both.
FIX	Ensure both services are on same network. Compose auto-creates default network if none defined.
	<pre>docker compose exec serviceA curl http://serviceB:8080/health</pre>

#66 docker compose ps Shows Unhealthy

SYMPTOM	Service listed as (unhealthy) in docker compose ps output.
ROOT CAUSE	Container healthcheck command failing — wrong endpoint, missing tool, or app not ready.
FIX	Test healthcheck manually inside container. Adjust interval/timeout/retries in healthcheck spec.
	<pre>docker compose exec <service> wget -qO- localhost:8080/health</pre>

#67 Long Shutdown Time in docker compose down

SYMPTOM	docker compose down takes 10+ seconds per service to stop.
ROOT CAUSE	Services not handling SIGTERM; Docker waits 10s timeout then SIGKILL.
FIX	Add signal handling in app. Or reduce stop_grace_period in Compose service config.
	<pre>stop_grace_period: 5s</pre>

#68 Compose Override File Not Applied

SYMPTOM	docker-compose.override.yml settings not reflected in running containers.
ROOT CAUSE	Using docker compose (v2 CLI) — override file should be docker-compose.override.yml in same dir.
FIX	Verify filename. Or explicitly specify: docker compose -f docker-compose.yml -f docker-compose.override.yml up.
	<pre>docker compose -f docker-compose.yml -f docker-compose.override.yml up -d</pre>

#69 Cannot Connect to Compose DB From Host IDE

SYMPTOM	IDE database client cannot connect to Compose DB container.
ROOT CAUSE	DB port not published to host in Compose config.
FIX	Add ports: - '5432:5432' to DB service in Compose file for development.
	<pre>docker compose port db 5432</pre>

#70 Compose Profiles Not Activating Services

SYMPTOM	Service with profile defined not starting with docker compose up.
ROOT CAUSE	--profile flag not passed to docker compose up.
FIX	Use docker compose --profile debug up to activate profile-specific services.
	<pre>docker compose --profile debug up -d</pre>

REGISTRY & SECURITY — Real-World Troubleshooting

#71 Private Registry Authentication Failing

SYMPTOM	docker push/pull fails: unauthorized: authentication required.
ROOT CAUSE	Not logged in to private registry, or credentials expired.
FIX	docker login <registry-url>. For CI, use --password-stdin to avoid credential in shell history.
	<pre>echo \$CI_TOKEN docker login registry.example.com -u \$CI_USER --password-stdin</pre>

#72 TLS Certificate Error Pushing to Private Registry

SYMPTOM	docker push fails: x509: certificate signed by unknown authority.
ROOT CAUSE	Registry uses self-signed cert; Docker doesn't trust it by default.
FIX	Add registry cert to /etc/docker/certs.d/<registry>/ca.crt. Or add to insecure-registries (dev only).
	<pre>sudo mkdir -p /etc/docker/certs.d/registry.example.com && sudo cp ca.crt /etc/docker/certs.d/registry.example.com/</pre>

#73 Image Vulnerability Scan Showing Critical CVEs

SYMPTOM	Security scan (Trivy/Snyk/Clair) shows critical CVEs in base image.
ROOT CAUSE	Outdated base image with known vulnerabilities in OS packages.
FIX	Rebuild with latest base image tag. Use distroless or minimal images. Automate weekly rebuilds.
	<pre>trivy image --severity CRITICAL,HIGH my-image:latest</pre>

#74 Docker Content Trust Signature Verification Failed

SYMPTOM	docker pull fails: Error: remote trust data does not exist or signature mismatch.
ROOT CAUSE	DOCKER_CONTENT_TRUST=1 set but image not signed, or wrong signer key.
FIX	Sign image with docker trust sign, or disable DCT for unsigned images in dev.
	<pre>docker trust sign myrepo/myimage:tag</pre>

#75 Container Running as Root Security Risk

SYMPTOM	Audit finds containers running as root; violates security policy.
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ROOT CAUSE	No USER instruction in Dockerfile; default Docker behavior is UID 0.
FIX	Add non-root USER in Dockerfile. Use --user flag. Enable userns-remap in daemon.
	<pre>RUN groupadd -r appgroup && useradd -r -g appgroup appuser USER appuser</pre>

#76 Privileged Container Security Concern

SYMPTOM	Container requires --privileged flag; security team flags as high risk.
ROOT CAUSE	App needs kernel capabilities; --privileged grants all caps + device access.
FIX	Use --cap-add for only needed capabilities (e.g., NET_ADMIN) instead of full --privileged.
	<pre>docker run --cap-add=NET_ADMIN --cap-drop=ALL my-image</pre>

#77 Docker Socket Mounted in Container (Security Risk)

SYMPTOM	Container has access to /var/run/docker.sock — container escape risk.
ROOT CAUSE	-v /var/run/docker.sock:/var/run/docker.sock mounted for Docker-in-Docker or CI agent.
FIX	Use Docker-outside-of-Docker pattern carefully. Prefer rootless Docker or Kaniko for CI builds.
	<pre>docker run --rm -v /var/run/docker.sock:/var/run/docker.sock -it docker:dind</pre>

#78 Registry Mirror Not Being Used

SYMPTOM	Pulls still going directly to Docker Hub despite mirror configured.
ROOT CAUSE	daemon.json registry-mirrors not applied (daemon not restarted), or per-image pull-through not set.
FIX	Restart Docker after updating daemon.json. Verify: docker info grep Mirror.
	<pre>docker info grep -i mirror</pre>

#79 Image Push Fails — Layer Already Exists But Digest Mismatch

SYMPTOM	docker push fails part-way: layer already exists but then checksum mismatch error.
ROOT CAUSE	Corrupted local layer cache or concurrent push conflict.
FIX	docker pull fresh, rebuild image cleanly. Clear builder cache: docker builder prune.
	<pre>docker builder prune -af && docker build --no-cache -t my-image . && docker push my-image</pre>

#80 Credentials Stored in Plain Text in ~/.docker/config.json

SYMPTOM	Security audit finds registry passwords in plain text in config.json.
ROOT CAUSE	No credential store configured; Docker defaults to base64 in config.json (not encrypted).
FIX	Install docker-credential-helpers (pass, secretservice). Configure credsStore in config.json.
<pre>sudo apt-get install pass && docker-credential-pass initialize</pre>	

RESOURCES & PERFORMANCE — Real-World Troubleshooting

#81 Container Using 100% CPU

SYMPTOM	Host CPU pegged; docker stats shows one container at 100% CPU.
ROOT CAUSE	Application in infinite loop, runaway process, or no CPU limits set.
FIX	Set --cpus=1.0 limit. Profile app. Use docker stats for monitoring.
	<pre>docker run --cpus=1.0 --memory=512m my-image</pre>

#82 Container OOM Killed by Kernel

SYMPTOM	Container exits suddenly; docker inspect shows OOMKilled: true.
ROOT CAUSE	Container exceeded memory limit; kernel OOM killer terminates it.
FIX	Increase --memory limit or fix memory leak in application.
	<pre>docker inspect <container> --format '{{.State.OOMKilled}}'</pre>

#83 Container Throttled Due to CPU Limits

SYMPTOM	App performs slower inside container than expected; CPU throttling metrics high.
ROOT CAUSE	--cpus set too low; CPU quota exceeded; CFS throttling kicking in.
FIX	Increase --cpus. Monitor: docker stats. Use --cpu-shares for softer limits.
	<pre>docker stats --no-stream --format 'table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}'</pre>

#84 docker stats Showing 0% CPU for All Containers

SYMPTOM	docker stats --no-stream shows 0.00% CPU for all containers unexpectedly.
ROOT CAUSE	cgroup v2 not supported by Docker version; or stats collected at wrong interval.
FIX	Upgrade Docker. Check cgroup version: stat -fc %T /sys/fs/cgroup/. Enable cgroup v2.
	<pre>docker stats --format 'table {{.Name}}\t{{.CPUPerc}}\t{{.MemPerc}}'</pre>

#85 Container Startup Slow Due to Large Image

SYMPTOM	New container takes 30+ seconds to become available after docker run.
ROOT CAUSE	Large image with many layers being pulled/extracted; or slow storage driver.

FIX	Optimize image size. Pre-pull image on host. Use image streaming or lazy loading registries.
	<code>docker pull my-image:latest && time docker run --rm my-image echo ready</code>

#86 High Memory Usage From Docker Daemon Itself

SYMPTOM	dockerd process consuming 1GB+ RAM on host.
ROOT CAUSE	Many containers/images, high event stream activity, or memory leak in older daemon.
FIX	Upgrade Docker daemon. Reduce number of running containers. Use docker system prune.
	<code>ps aux grep dockerd && docker system df</code>

#87 I/O Throttling Needed for Container

SYMPTOM	Container doing heavy disk I/O starving other containers or host processes.
ROOT CAUSE	No I/O limits set; container uses full disk bandwidth.
FIX	Use --device-read-bps and --device-write-bps to throttle container disk I/O.
	<code>docker run --device-write-bps /dev/sda:10mb --device-read-bps /dev/sda:10mb my-image</code>

#88 Container Fork Bomb Crashing Host

SYMPTOM	Host becomes unresponsive; too many processes spawned from container.
ROOT CAUSE	No PID limit set; container process spawns unlimited child processes.
FIX	Set --pids-limit to restrict number of processes a container can create.
	<code>docker run --pids-limit=100 my-image</code>

#89 Network Throughput Limited in Container

SYMPTOM	Container network throughput much lower than host; iperf shows degraded bandwidth.
ROOT CAUSE	Bridge network overhead; or tc qdisc rules limiting container bandwidth.
FIX	Use --network=host for performance-critical workloads. Or tune MTU settings.
	<code>docker run --network=host iperf3 -c <server></code>

#90 Huge Number of Zombie Containers Slowing Daemon

SYMPTOM	docker ps -a shows hundreds of Exited containers; daemon becomes sluggish.
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ROOT CAUSE	Containers run without --rm flag accumulate as Exited state.
FIX	Use --rm for ephemeral containers. Periodic cleanup: docker container prune.
	<pre>docker container prune -f && docker ps -a wc -l</pre>

ADVANCED & DEBUGGING — Real-World Troubleshooting

#91 Docker Daemon Not Starting After Config Change

SYMPTOM	<code>sudo systemctl start docker</code> fails; daemon exits immediately.
ROOT CAUSE	Syntax error in <code>/etc/docker/daemon.json</code> .
FIX	Validate JSON: <code>python3 -m json.tool /etc/docker/daemon.json</code> . Fix error and restart.
	<pre>python3 -m json.tool /etc/docker/daemon.json && sudo systemctl restart docker</pre>

#92 BuildKit Not Enabled — Advanced Build Features Missing

SYMPTOM	<code>--mount=type=cache</code> and <code>--secret mount</code> in Dockerfile not working.
ROOT CAUSE	BuildKit not enabled; classic builder doesn't support these features.
FIX	Set <code>DOCKER_BUILDKIT=1</code> env var, or enable in <code>daemon.json</code> : <code>features.buildkit: true</code> .
	<pre>DOCKER_BUILDKIT=1 docker build -t my-image .</pre>

#93 docker build Fails in Corporate Proxy Environment

SYMPTOM	<code>RUN apt-get install</code> or <code>pip install</code> fails: could not connect to server.
ROOT CAUSE	Container doesn't inherit host proxy settings during build.
FIX	Pass proxy as build args or set in <code>daemon.json</code> <code>httpProxy/httpsProxy</code> .
	<pre>docker build --build-arg HTTP_PROXY=\$HTTP_PROXY --build-arg HTTPS_PROXY=\$HTTPS_PROXY .</pre>

#94 Rootless Docker Containers Cannot Bind to Port < 1024

SYMPTOM	Rootless Docker container fails: permission denied binding to port 80 or 443.
ROOT CAUSE	Rootless mode runs as non-root user; binding ports < 1024 requires privilege.
FIX	Use port >= 1024 and reverse proxy (Nginx/Caddy). Or: <code>sysctl net.ipv4.ip_unprivileged_port_start=80</code> .
	<pre>sudo sysctl -w net.ipv4.ip_unprivileged_port_start=80</pre>

#95 Container Cannot Access GPU

SYMPTOM	Machine learning container cannot detect GPU; <code>nvidia-smi</code> returns command not found.
ROOT CAUSE	NVIDIA Container Toolkit not installed, or <code>--gpus</code> flag not used.

FIX	Install nvidia-container-toolkit. Run with --gpu all flag.
	<code>docker run --gpu all nvidia/cuda:11.0-base nvidia-smi</code>

#96 Docker Events Stream Shows Unexpected Container Kills

SYMPTOM	Containers dying without obvious reason; docker events shows die events.
ROOT CAUSE	Health check failing causing restart policy to kill, or resource limit hit, or external signal.
FIX	docker events --filter container=<id> --since 1h. Check OOM and health status.
	<code>docker events --filter container=<id> --since 1h</code>

#97 Cannot Copy File Into Running Container

SYMPTOM	docker cp returns: No such container:path or permission denied.
ROOT CAUSE	Target path doesn't exist in container, or container filesystem is read-only.
FIX	Ensure path exists: docker exec mkdir -p /target. Check if container uses read-only rootfs.
	<code>docker cp ./localfile.txt <container>:/app/localfile.txt</code>

#98 docker inspect Returns Unexpected Network Config

SYMPTOM	Container IP and network settings different from what was configured.
ROOT CAUSE	Default bridge network used instead of custom network; Compose network naming.
FIX	Explicitly specify network at run time. Use docker inspect to verify.
	<code>docker inspect <container> --format='{{json .NetworkSettings.Networks}}' python3 -m json.tool</code>

#99 Container Filesystem Changes Not Persisted

SYMPTOM	Changes made inside running container lost after container stop/start.
ROOT CAUSE	Writes to container writable layer — not persisted across container lifecycle without volumes.
FIX	Mount a volume for any data that must persist. Use docker commit only for debugging (not production).
	<code>docker run -v /host/path:/app/data -it my-image</code>

#100 Docker Daemon TLS Configuration Breaking Remote Access

SYMPTOM	Remote docker client gets: tls: bad certificate or connection refused on port 2376.
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ROOT CAUSE	TLS certs misconfigured for Docker daemon remote access; cert SANs don't match server hostname.
FIX	Regenerate TLS certs with correct server hostname/IP in SAN. Configure DOCKER_HOST and DOCKER_TLS_VERIFY.
	<pre>export DOCKER_HOST=tcp://<host>:2376 && export DOCKER_TLS_VERIFY=1 && docker info</pre>